

LESSON 8-5 • SESSION 1 • GUIDED NOTES

Graphing Inequalities

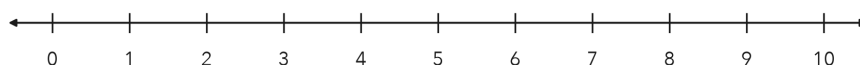
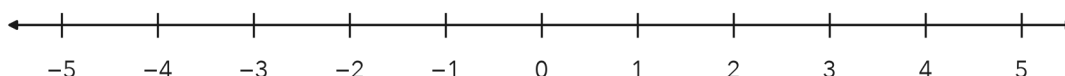
Learning target – I can graph an inequality on a number line using an open or a closed dot. · 6.EE.B.8

WORDS TO KNOW*fill in the missing words*

WORD	WHAT IT MEANS	EXAMPLE
boundary point <i>punto límite</i>	The number where the solutions _____ or stop.	the 8 in $x \geq 8$
closed circle <i>círculo cerrado</i>	A filled dot: the boundary ____ a solution.	used for $\leq$ and $\geq$
open circle <i>círculo abierto</i>	An empty dot: the boundary is ____ a solution.	used for $<$ and $>$
solution set <i>conjunto solución</i>	_____ the numbers that make an inequality true.	$x \geq 8$: 8, 9.5, 20, ...

KEY IDEA*complete each sentence*

- Use a _____ dot for $\leq$ and $\geq$. Use an _____ dot for $<$ and $>$.
- Draw the arrow toward the numbers that make the inequality _____.

WORKED EXAMPLE*fill in the blanks as we solve*Graph $x < 6$ on a number line.**Step 1** The boundary number is ____.**Step 2** $<$ does not include 6, so draw an _____ dot at 6.**Step 3** x is less than 6, so the arrow points _____.**Answer** **Check:** $5 < 6$ is _____, and 5 is under the arrow.**YOU TRY***use the example as a model***1.** Graph $y \geq -2$ on the number line.**2.** Write the inequality for each graph, using x .**a.** closed dot at 4, arrow left _____**b.** open dot at -1 , arrow right _____**SUM IT UP***in your own words*

I choose an open or a closed dot by looking at _____, and I point the arrow _____.

LESSON 8-5 • SESSION 2 • GUIDED NOTES**Testing Solutions to Inequalities**

Learning target – I can test values against an inequality and graph the solution set. • 6.EE.B.8

WORDS TO KNOW*fill in the missing words*

WORD	WHAT IT MEANS	EXAMPLE
substitute <i>sustituir</i>	Put a number in _____ of the variable.	$x = 9$ in $x > 5$ gives $9 > 5$
solution <i>solución</i>	A value that makes the inequality _____.	9 is a solution of $x > 5$
true statement <i>enunciado verdadero</i>	A comparison that is _____.	$9 > 5$

KEY IDEA*complete each sentence*

- To test a value, substitute it and ask whether the statement is _____.
- For $\leq$ and $\geq$, the boundary _____ a solution. For $<$ and $>$, it is _____.

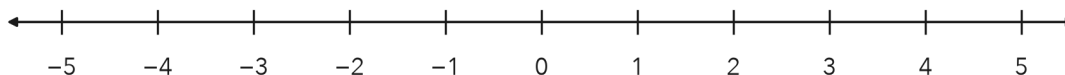
WORKED EXAMPLE*fill in the blanks as we solve*Test 8, 9 and 10 in $x < 9$.**Step 1** $8 < 9$ is _____, so 8 is a solution.**Step 2** $9 < 9$ is _____, so the dot at 9 is _____.**Step 3** $10 < 9$ is false, so 10 is not a solution.**Answer** Only _____ works. **Graph: open dot at 9, arrow left.****YOU TRY***use the example as a model*

1. Is the value a solution? Write yes or no.

a. $y \geq 3$; $y = 3$ _____

b. $m < -2$; $m = 0$ _____

2. Test -1 , 2 and 5 in $x > 2$. Then graph $x > 2$.



Solutions _____

SUM IT UP*in your own words*

To decide whether a number is a solution, I _____ and the boundary is a solution only when _____.
